

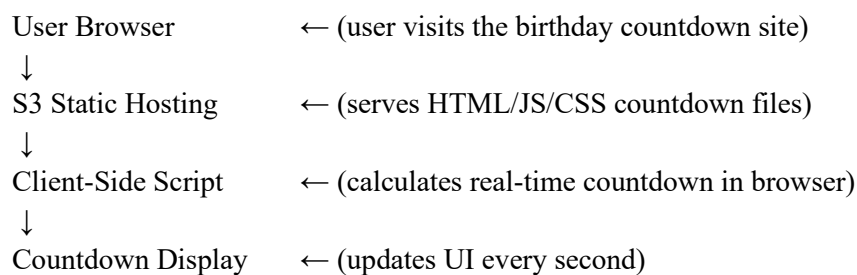
Serverless Birthday Countdown website using AWS S3

STATIC WEBSITE HOSTING ON AMAZON S3:

~ project-name: **s3-static-bday-countdown**

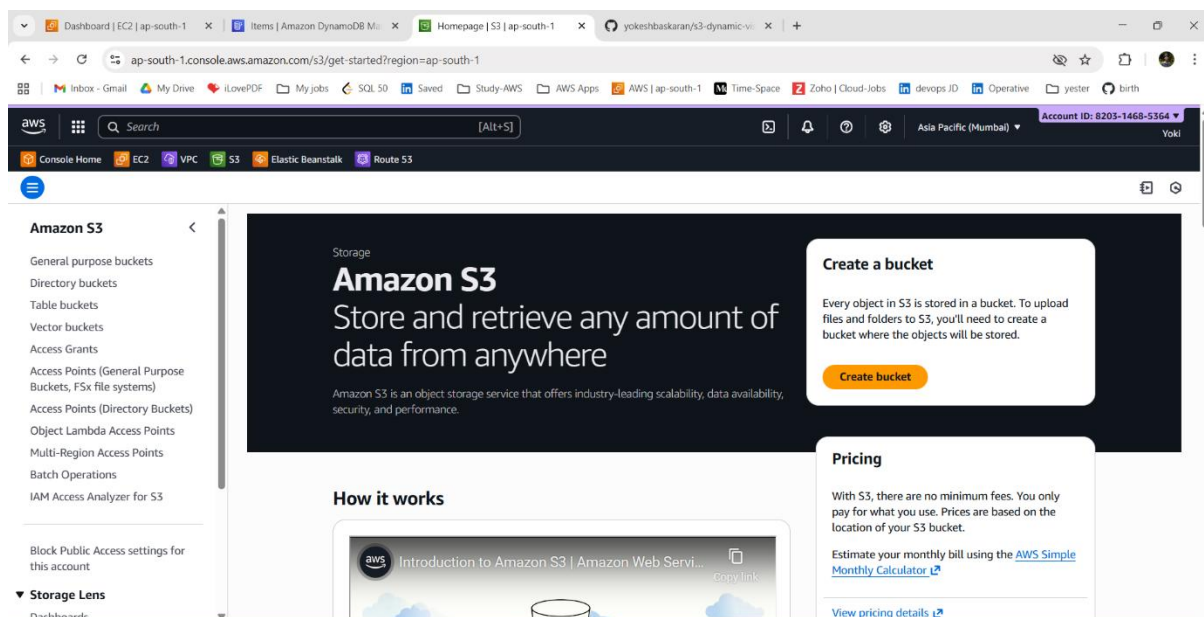
- This document describes how to host a static website using Amazon S3's built-in static hosting feature.
- This approach is suitable for simple static sites that do not require HTTPS, backend logic, authentication, or private hosting.

ARCHITECTURE DIAGRAM:



S3 STATIC HOSTING:

- S3 static hosting in AWS is a feature that lets you host static websites (HTML, CSS, JavaScript, images) directly from an Amazon S3 bucket without needing a web server.
- It serves your site via HTTP/HTTPS with high availability and low cost.



Here, we will see how to host a static website using Amazon S3's built-in static hosting feature. This approach is suitable for simple static sites that do not require HTTPS, backend logic, authentication, or private hosting.

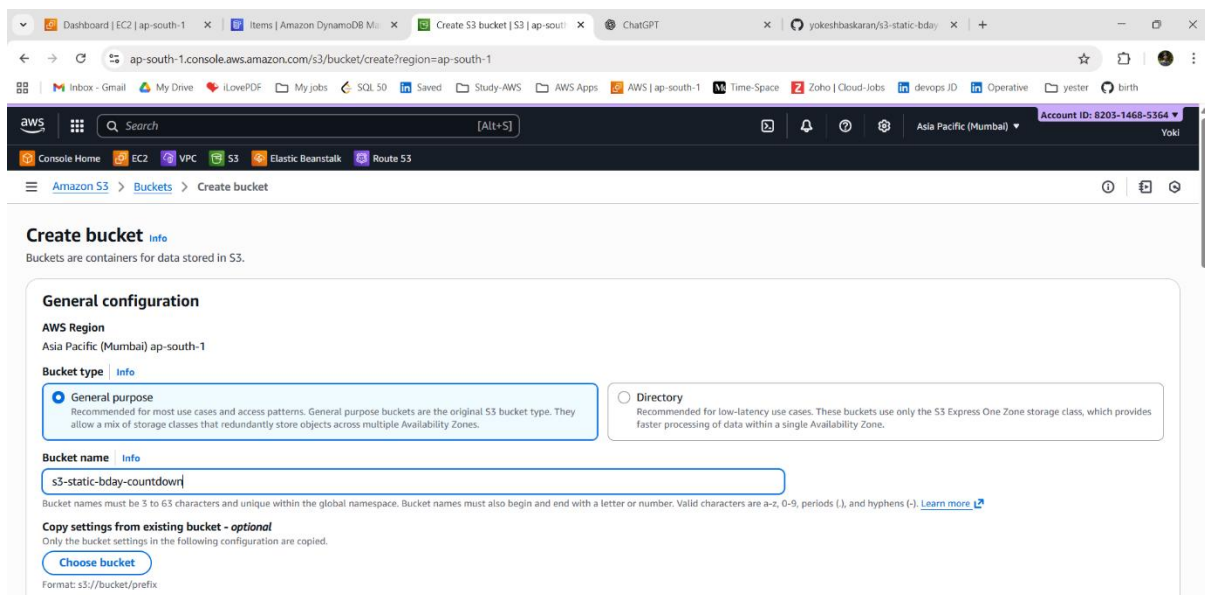
~ Go → AWS Management Console and Open the S3 service.

1. Create an S3 Bucket:

i) Choose “Create bucket”.

ii) Enter the bucket name (must be globally unique).

Bucket-name: **s3-static-bday-countdown**



Dashboard | EC2 | ap-south-1 | Items | Amazon DynamoDB M... | Create S3 bucket | S3 | ap-south-1 | ChatGPT | yokeshbaskaran/s3-static-bday | +

ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1

Search [Alt+S]

Console Home EC2 VPC S3 Elastic Beanstalk Route 53

Amazon S3 > Buckets > Create bucket

Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
Asia Pacific (Mumbai) ap-south-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

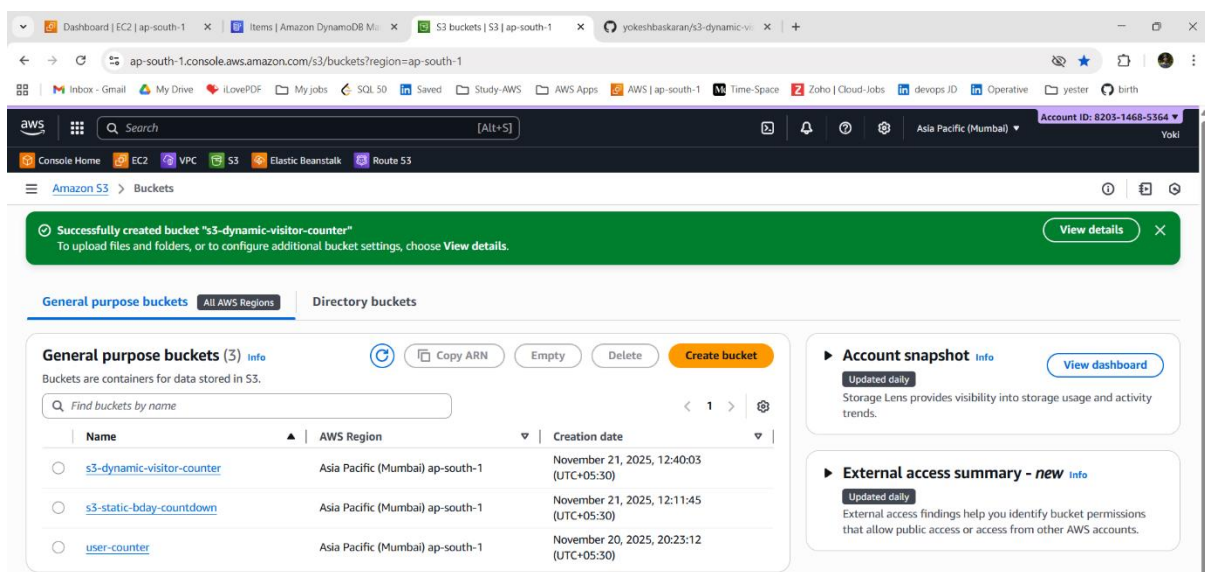
☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [info](#)
s3-static-bday-countdown
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
[Choose bucket](#)
Format: s3://bucket/prefix

iii) Select an AWS Region - **ap-south-1**

iv) Create the bucket.



Dashboard | EC2 | ap-south-1 | Items | Amazon DynamoDB M... | S3 buckets | S3 | ap-south-1 | yokeshbaskaran/s3-dynamic-vi... | +

ap-south-1.console.aws.amazon.com/s3/buckets?region=ap-south-1

Search [Alt+S]

Console Home EC2 VPC S3 Elastic Beanstalk Route 53

Amazon S3 > Buckets

Successfully created bucket "s3-dynamic-visitor-counter"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

General purpose buckets [All AWS Regions](#) Directory buckets

General purpose buckets (3) [info](#)

Buckets are containers for data stored in S3.

[Find buckets by name](#)

Name	AWS Region	Creation date
s3-dynamic-visitor-counter	Asia Pacific (Mumbai) ap-south-1	November 21, 2025, 12:40:03 (UTC+05:30)
s3-static-bday-countdown	Asia Pacific (Mumbai) ap-south-1	November 21, 2025, 12:11:45 (UTC+05:30)
user-counter	Asia Pacific (Mumbai) ap-south-1	November 20, 2025, 20:23:12 (UTC+05:30)

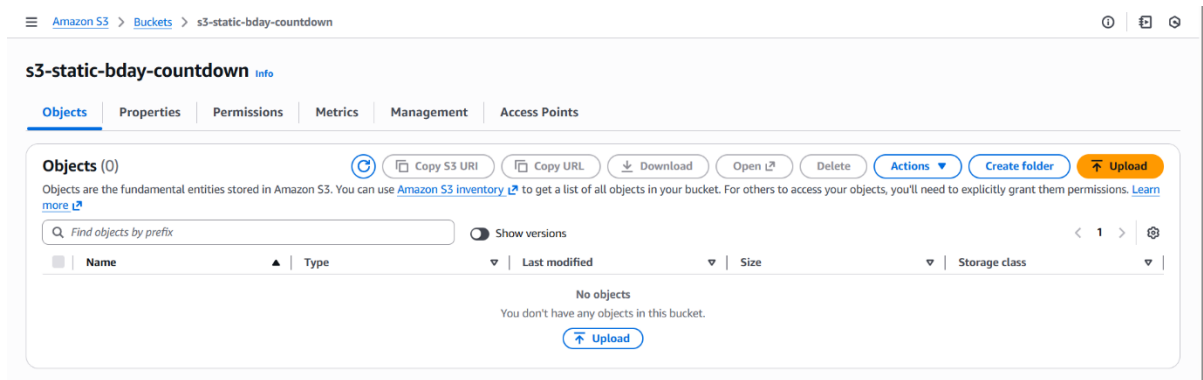
[Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Account snapshot [info](#) [View dashboard](#)
[Updated daily](#)
Storage Lens provides visibility into storage usage and activity trends.

External access summary - new [info](#)
[Updated daily](#)
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

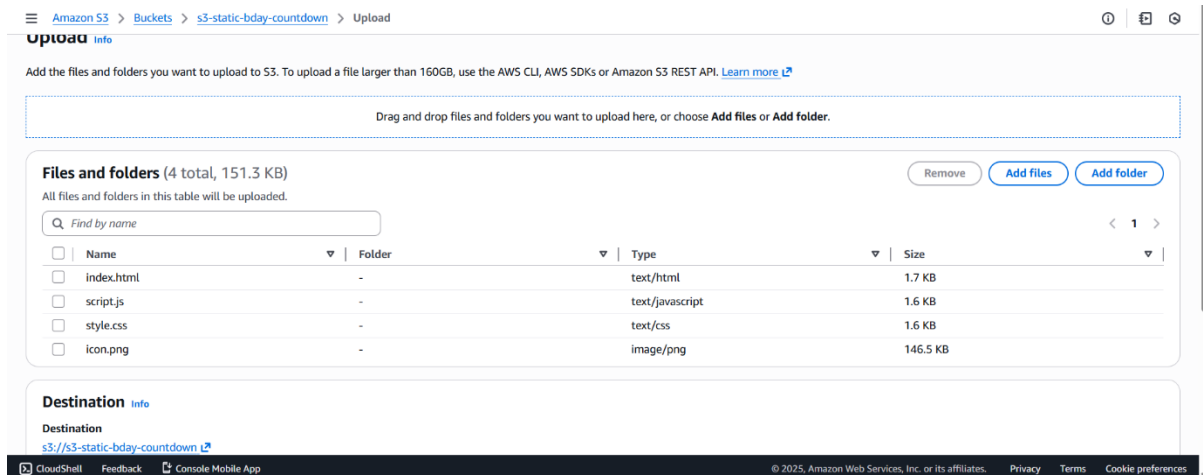
2. Upload Your Website Files:

i) Open the **s3-static-bday-countdown** bucket.

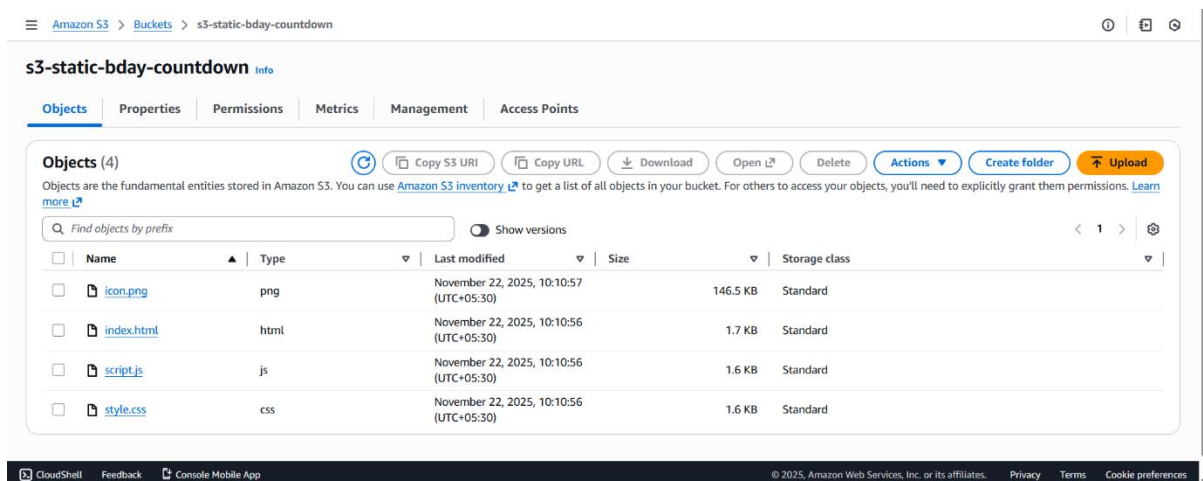


ii) Choose **Upload** to upload the files.

iii) Add all files related to your website (e.g., index.html, error.html, CSS, JS files).

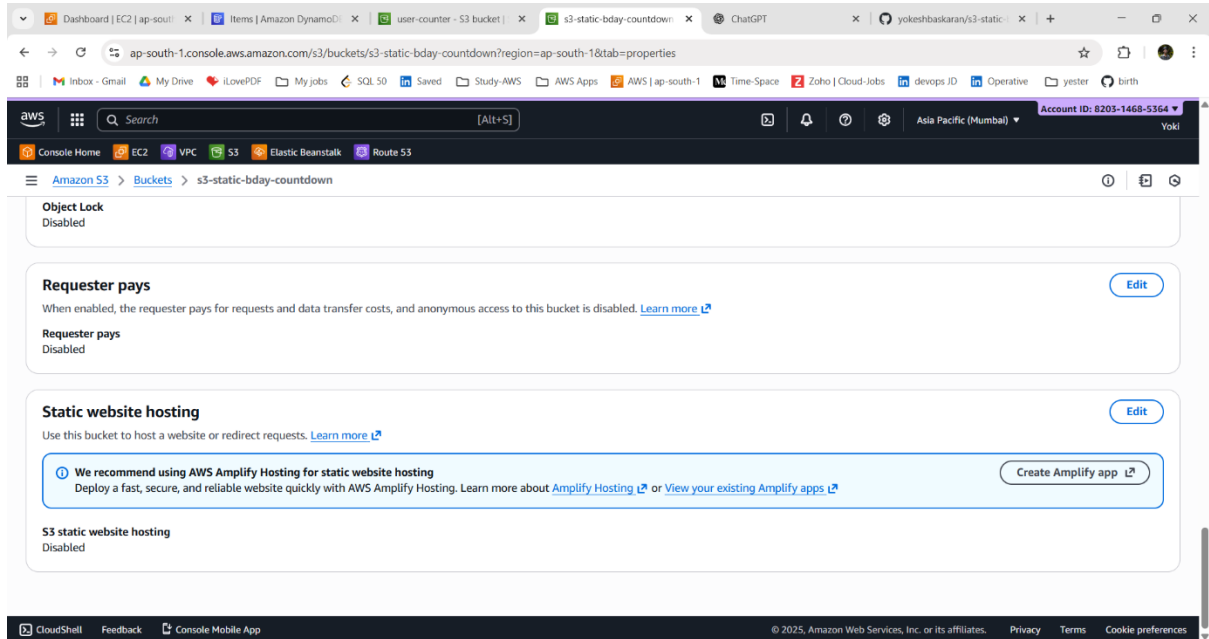


iv) Choose **Upload** again to complete the process.

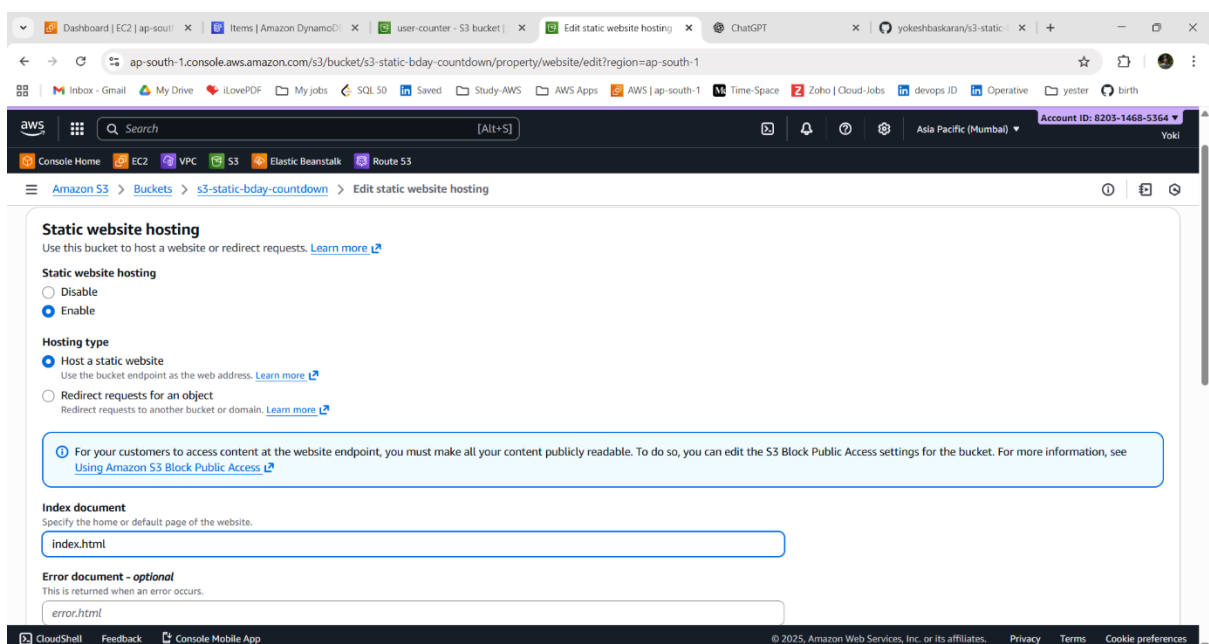


3. Enable Static Website Hosting:

- i) Go to the “**Properties**” tab of the bucket.
- ii) Scroll to **Static website hosting**.
- iii) Choose **Edit** and then Select **Enable**.

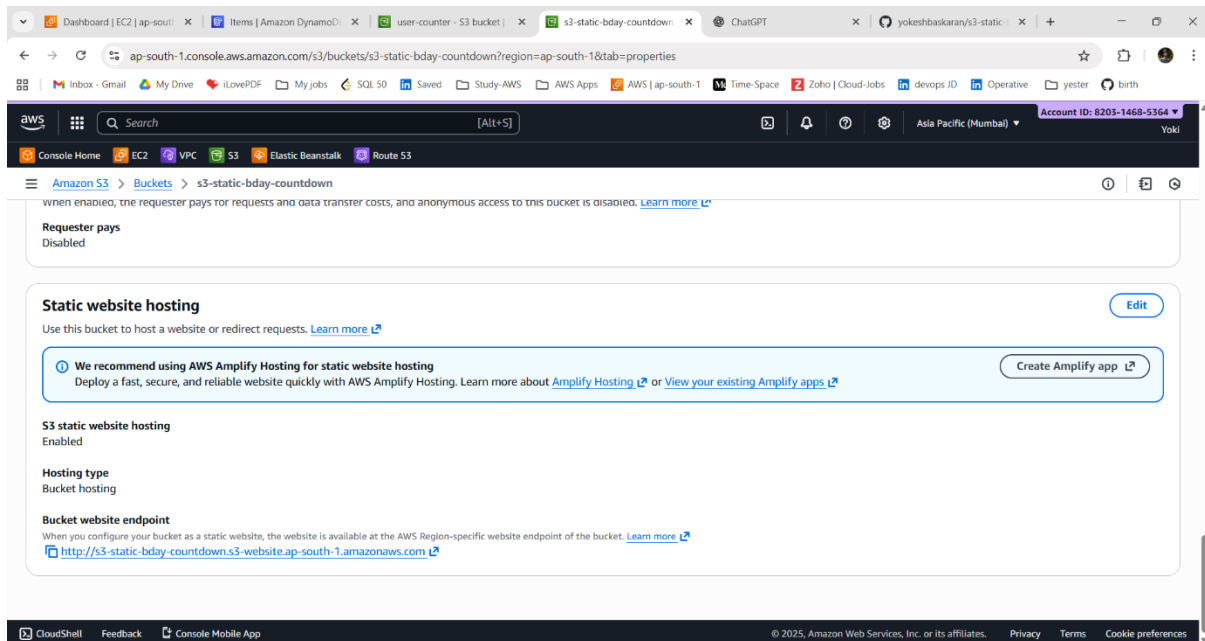


- iv) Choose Host a static website.
- v) Set your Index document (e.g., **index.html**).
- vi) click **Save changes**.



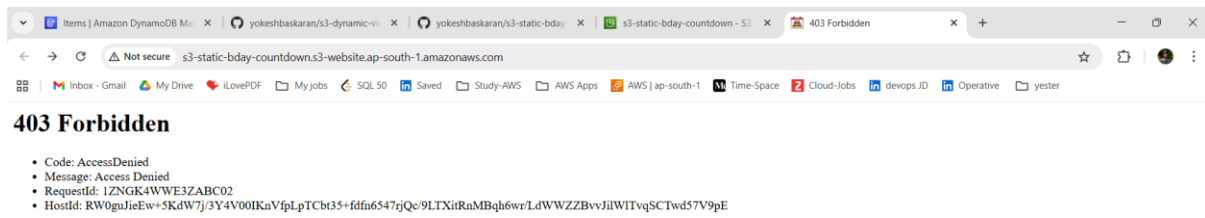
vii) Copy the Website endpoint shown after saving — this is your public website URL.

URL: `http://s3-static-bday-countdown.s3-website.ap-south-1.amazonaws.com`



ERROR:

~ If we get 403 Forbidden or access denied. We need to check the bucket settings.

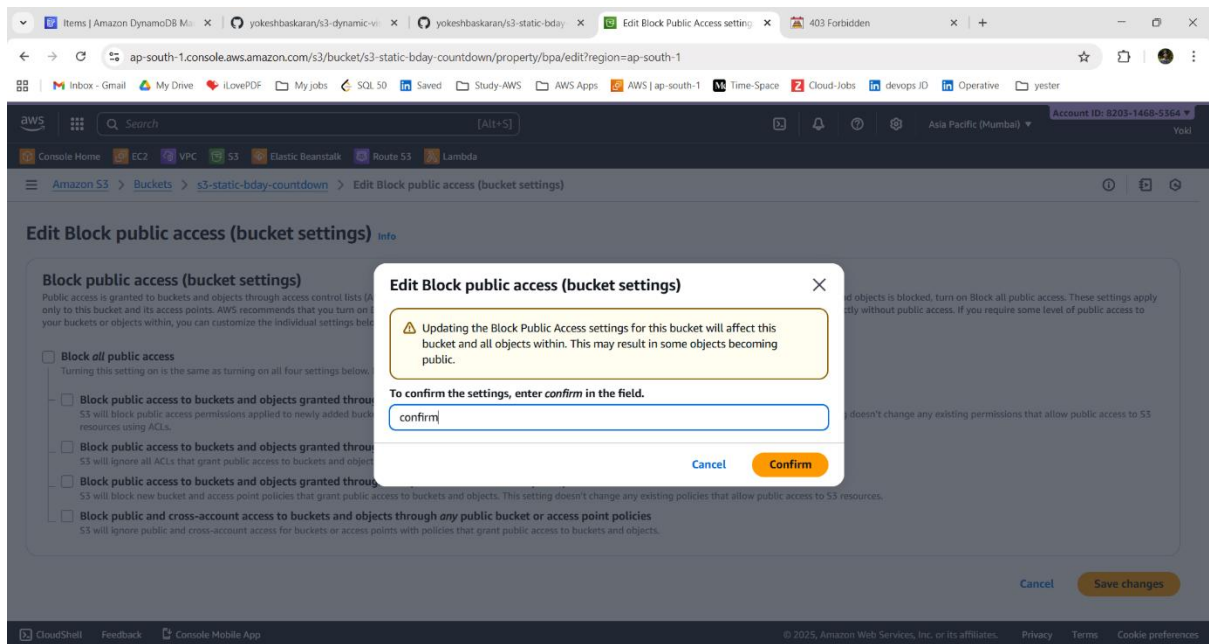


~ we need to **Turn off “Block Public Access”** (bucket settings).

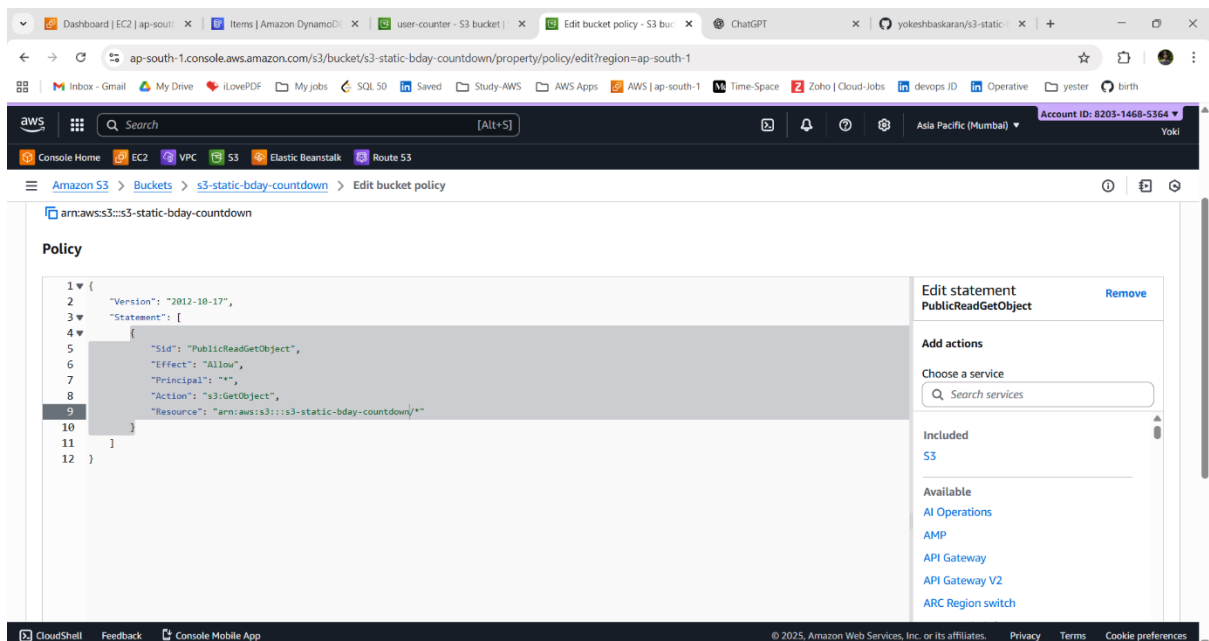
~ Also, give the **Bucket policy**.

4. Allow Public Access to Website Files and add Bucket Policy:

- i) Go to the “**Permissions**” tab of the bucket.
- ii) Scroll to “**Block Public Access**” (bucket settings).
- iii) Choose **Edit** and **uncheck** the “Block all Public Access”.
- iv) type “confirm” and click Save Changes.



- v) In same page of “**Permissions**” tab of the bucket. Scroll to **Bucket policy**.
- vi) Choose **Edit** and add a policy that allows public read of all objects in the bucket.

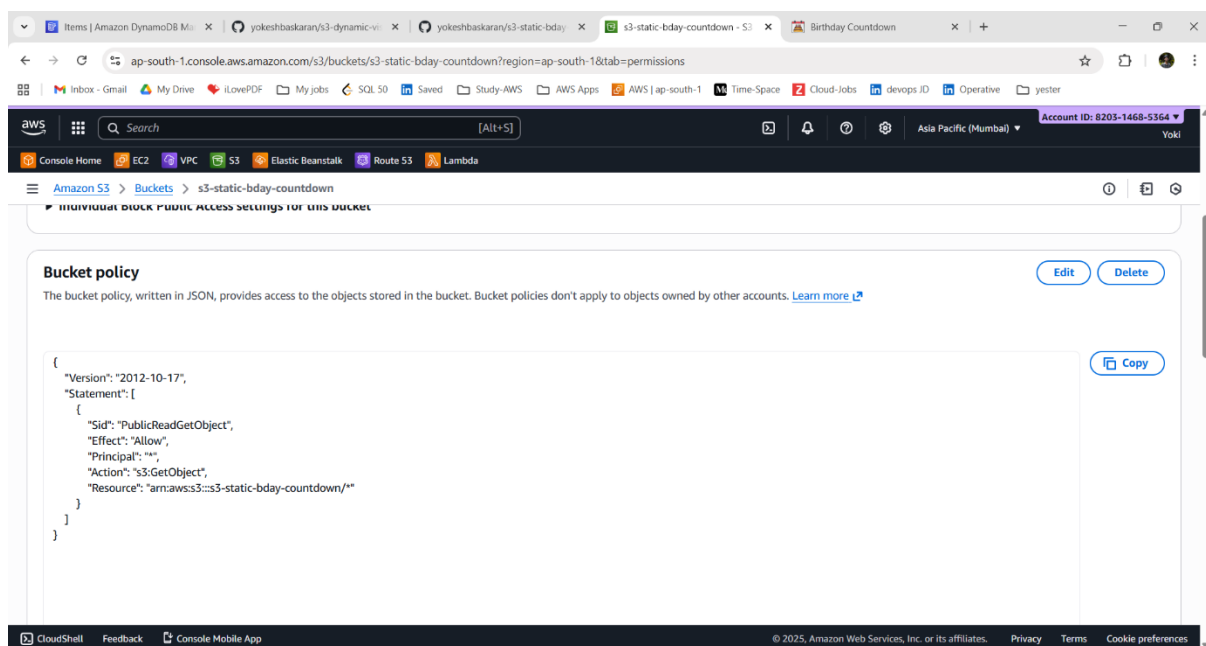


BUCKET POLICY:

```
'''  
  
{  
  
    "Version": "2012-10-17",  
  
    "Statement": [  
  
        {  
  
            "Sid": "PublicReadGetObject",  
  
            "Effect": "Allow",  
  
            "Principal": "*",  
  
            "Action": "s3:GetObject",  
  
            "Resource": "arn:aws:s3:::s3-static-bday-countdown/*"  
  
        }  
  
    ]  
  
}  
  
'''
```

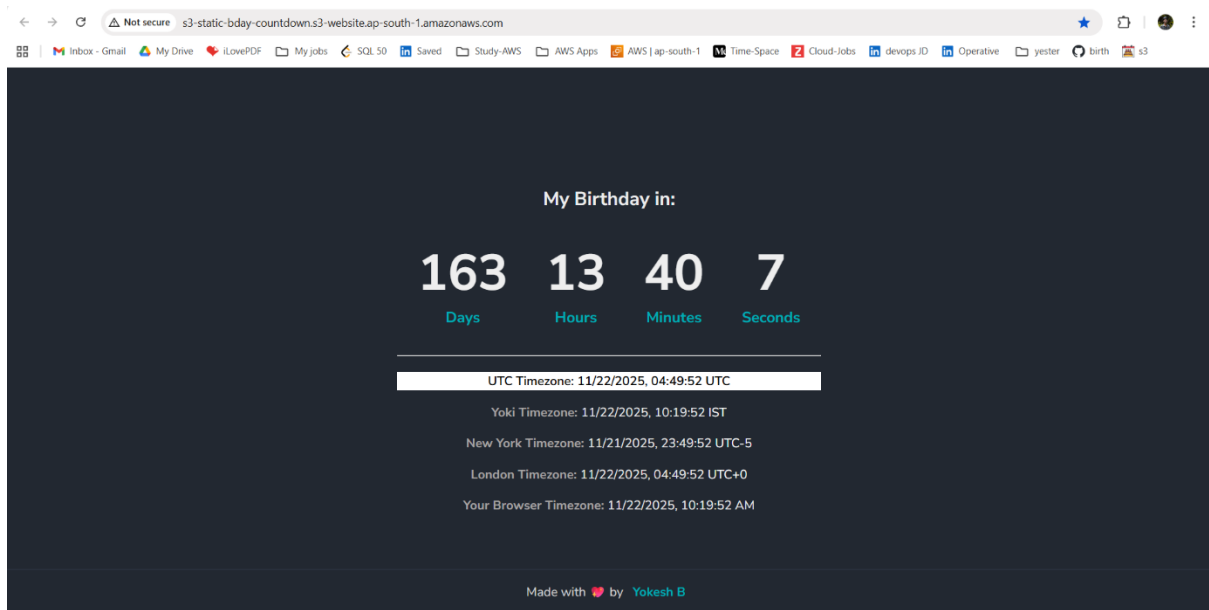
vii) Save the policy.

viii) Verify that Public Access indicators show that the bucket allows public reads.



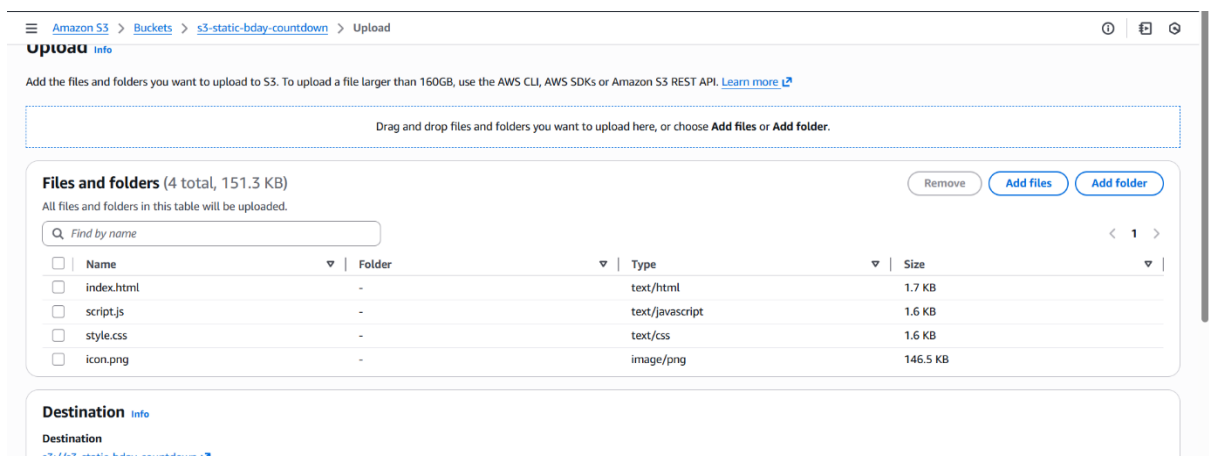
5. Verify Website Accessibility:

- i) Open the S3 Website endpoint URL in your browser.
- ii) Confirm that the site loads and that resources (CSS, JS files) work properly.
- iii) If something doesn't load, check:
 - Files are uploaded at the correct paths
 - Permissions allow public read
 - File names match exactly (case-sensitive)



6. Maintenance & Updates:

- i) To update your site, upload new versions of your files to the S3 bucket.
- ii) Overwrite existing objects or delete and re-upload.
- iii) Refresh the website to view changes (sometimes a hard refresh is needed: Ctrl+Shift+R).



~ This concludes the steps for hosting a static website on Amazon S3.

~ Your site is now ready to be accessed from anywhere in the world.